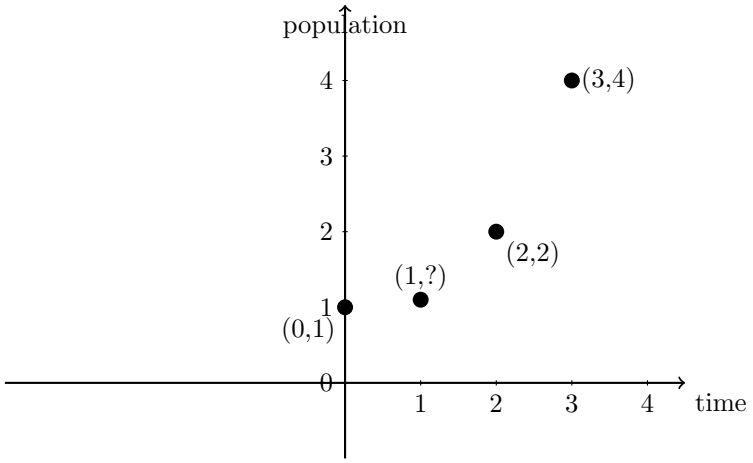
Numerical Differentiation

2024 Spring, UIC-AM-NUM, by Xiaofeng Cai

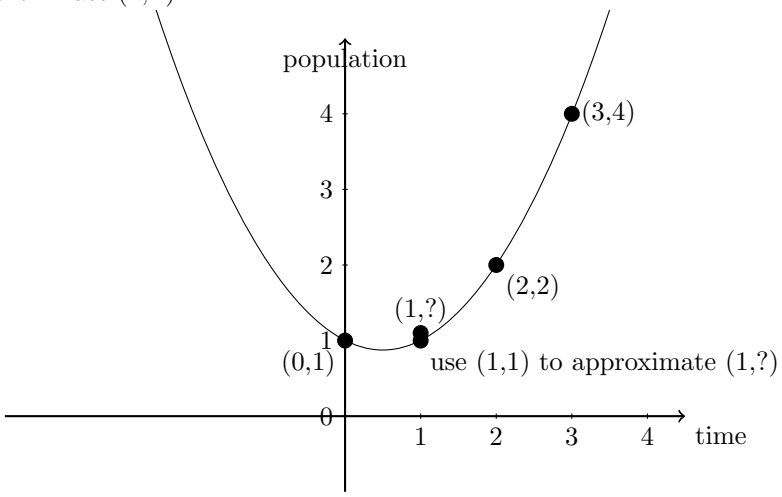
# 1 Motivation

We have a set of data that is the population of a commodity as a function of time. Let's draw a graph to present this data.




As you can see, we have recorded the data, $(0,1),(1,?),(2,2),(3,4)$ presented in the graph. We find that someone missed one data about year 1.

We know that we can use the interpolating polynomial by using these three points $(0,1),(2,2),(3,4)$ to approximate $(1,?)$




We denote the function of temperature with respect to time as $f(x)$. So we have the following table.

| x | 0 | 2 | 3 |
|---|---|---|---|
| f(x) | 1 | 2 | 4 |

We know we can use the interpolating polynomial $p(x)$ to approximate $f(1)$. A question arises:

**Question 1:** How to approximate $f'(1)$?

**Answer to Question 1:** We can approximate $p_2'(1)$ to approximate $f'(1)$. Such approximation is called **numerical differentiation**.

Before we study the new topic, let's briefly review Taylor's Theorem.

## 2 Review of Taylor's Theorem

**Taylor's Theorem** If $f(x)$ is $k$ times continuously differentiable, then

$$f(X) = f(a)+f'(a)(X-a)+\frac{f''(a)}{2!}(X-a)^2+\frac{f'''(a)}{3!}(X-a)^3+\cdots+\frac{f^{(k-1)}(a)}{(k-1)!}(X-a)^{k-1}+\frac{f^{(k)}(c)}{(k)!}(X-a)^k \tag{1}$$

where $c$ is between $a$ and $X$.

**For example:**

$$e^h = 1 + h + \frac{h^2}{2!} + \frac{h^3}{3!} + \cdots . \tag{2}$$

Then

$$e \approx 1 + 1 + \frac{1}{2!} + \cdots + \frac{1}{9!} = 2.71828. \tag{3}$$

Now let's develop numerical differentiation formulas for approximating derivatives.

## 3 Finite difference formulas

Consider the case when $n = 1$ so that

$$p_1(x) = f_0 + f[x_0, x_1](x - x_0). \tag{4}$$

Then

$$p_1'(x) = f[x_0, x_1] = \frac{f_1 - f_0}{x_1 - x_0}. \tag{5}$$

If we set $x_0 = x$ and $x_1 = x + h$, then $x_1 - x_0 = h$ and

**Two-point forward-difference formula**

$$f'(x) \approx \frac{f(x + h) - f(x)}{h} \equiv D_h^F(f). \tag{6}$$

How good is the this approximation? Let's estimate its error.

**The error of two-point forward-difference formula** The following difference is called the error $E$,

$$E = f'(x) - \frac{f(x + h) - f(x)}{h}. \tag{7}$$

This could be done using **Taylor's Theorem** in the following way.

Taylor's Theorem says that if $f$ is twice continuously differentiable, then

$$f(x + h) = f(x) + hf'(x) + \frac{h^2}{2}f''(c), \tag{8}$$

where $c$ is between $x$ and $x + h$. Then

$$\frac{f(x + h) - f(x)}{h} = f'(x) + \frac{h}{2}f''(c), \tag{9}$$

so that

$$E = -\frac{h}{2}f''(c). \tag{10}$$

If, for every $t \in [a, b]$, $|f''(t)| \leq M_2$, where $M_2$ is a positive constant, then

$$|E| \leq \frac{1}{2}hM_2, \tag{11}$$

and we write

$$|E| = O(h) \tag{12}$$

. Roughly speaking, this means that if $h$ is halved, the error will be reduced by a factor of 2. Since $h$ appears in (12) raised to the power 1, the forward difference formula (6) is said to be **first order**.

**Example 1.** Using $D_h^F(f)$ of (6), approximate the derivative of the function

$$f(x) = e^x$$

at $x = 0$. Results for various values of $h$ are given in Table 1. Ratio is the quantity $\left|\frac{D_h^F(f)-e^0}{D_{h/2}^F(f)-e^0}\right|$ and shows that, for $h$ sufficiently small, the error is reduced by a factor of two when $h$ is halved.

Table 1: Example of numerical differentiation using the forward difference.

| $h$ | $D_h^F$ | Error | Ratio |
|---|---|---|---|
| 0.2000 | 1.1070 | 0.1070 | - |
| 0.1000 | 1.0517 | 0.0517 | 2.0695 |
| 0.0500 | 1.0254 | 0.0254 | 2.0340 |
| 0.0250 | 1.0126 | 0.0126 | 2.0168 |
| 0.0125 | 1.0063 | 0.0063 | 2.0084 |

Returning to (5) and setting $x_0 = x - h$ and $x_1 = x + h$, then $x_1 - x_0 = 2h$ and

**Three-point centered-difference formula**

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h} \equiv D_h^C(f). \tag{13}$$

How good is the this approximation? Let's estimate its error.

**The error of three-point centered-difference formula** The following difference is called the error $E$,

$$E = f'(x) - \frac{f(x+h) - f(x-h)}{2h}. \tag{14}$$

This could be done using **Taylor's Theorem** in the following way.

Taylor's Theorem says that if $f$ is three times continuously differentiable, then

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2}f''(x) + \frac{h^3}{6}f'''(c_1) \tag{15}$$

and

$$f(x-h) = f(x) - hf'(x) + \frac{h^2}{2}f''(x) - \frac{h^3}{6}f'''(c_2), \tag{16}$$

where $x - h < c_2 < x < c_1 < x + h$. Then

$$E = f'(x) - \frac{f(x+h) - f(x-h)}{2h} = -\frac{h^2}{6}\left(\frac{f'''(c_1) + f'''(c_2)}{2}\right) = -\frac{h^2}{6}f'''(c) \tag{17}$$

for some c between $c_1$ and $c_2$, on using the Intermediate Value Theorem. Thus, in this case,

$$E = O(h^2) \tag{18}$$

and the centred difference formula (13) is **second order**.

**Example 1.** Using $D_h^C(f)$ of (13), approximate the derivative of the function

$$f(x) = e^x$$

at $x = 0$. Results for various values of $h$ are given in Table 2. Ratio is the quantity $\left|\frac{D_h^C(f)-e^0}{D_{h/2}^C(f)-e^0}\right|$ and shows that, for $h$ sufficiently small, the error is reduced by a factor of four when $h$ is halved.

Table 2: Example of numerical differentiation using the centered difference.

| $h$ | $D_h^C$ | Error | Ratio |
|---|---|---|---|
| 0.2000 | 1.0067 | 0.0067 | - |
| 0.1000 | 1.0017 | 0.0017 | 4.0060 |
| 0.0500 | 1.0004 | 0.0004 | 4.0015 |
| 0.0250 | 1.0001 | 0.0001 | 4.0004 |
| 0.0125 | 1.0000 | 0.0000 | - |

To obtain an approximation to $f''(x)$, consider

$$p_2(x) = f(x_0) + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1), \tag{19}$$

the polynomial of degree $\leq 2$ interpolating $f(x)$ at $x_0, x_1$ and $x_2$. Then

$$p_2''(x) = 2f[x_0, x_1, x_2], \tag{20}$$

and if $x_0 = x - h, x_1 = x$ and $x_2 = x + h$, it follows that

$$p_2''(x) = \frac{f(x-h) - 2f(x) + f(x+h)}{h^2} \equiv D_h''f(x). \tag{21}$$

Using Taylor series expansions, it can be shown that

$$f''(x) - D_h''f(x) = -\frac{h^2}{12}f^{(4)}(c), \tag{22}$$

where $c$ lies between $x - h$ and $x + h$.

So far, we know the method of constructing the interpolating polynomial to derive a numerical differentiation formula. Let's do the following example.

**Example 2.** Develop a second-order formula for the first derivative $f'(x)$ in terms of $f(x), f(x-h)$ and $f(x-2h)$ by constructing the interpolating polynomial.

**Solution:** Given three points $(x_0, f(x_0)), (x_1, f(x_1))$ and $(x_2, f(x_2))$, then we have the interpolating polynomial

$$p_2(x) = f(x_0) + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1), \tag{23}$$

the polynomial of degree $\leq 2$ interpolating $f(x)$ at $x_0, x_1$ and $x_2$. Then

$$\begin{aligned} p_2'(x) &= f[x_0, x_1] + f[x_0, x_1, x_2](2x - x_0 - x_1) \\ &= \frac{f(x_1) - f(x_0)}{x_1 - x_0} + \frac{\frac{f(x_2)-f(x_1)}{x_2-x_1} - \frac{f(x_1)-f(x_0)}{x_1-x_0}}{x_2 - x_0}(2x - x_0 - x_1), \end{aligned}$$

and if $x_0 = x$, $x_1 = x - h$, and $x_2 = x - 2h$, it follows that

$$\begin{aligned} p_2'(x) &= \frac{f(x-h) - f(x)}{-h} + \frac{\frac{f(x-2h)-f(x-h)}{x-2h-(x-h)} - \frac{f(x-h)-f(x)}{x-h-x}}{x - 2h - x}(2x - x - (x-h)) \\ &= \frac{f(x-2h) - 4f(x-h) + 3f(x)}{2h}. \end{aligned}$$

Now we can use Taylor's Theorem to verify the second order of the formula. According to Taylor's Theorem, if $f$ is three times continuously differentiable, then

$$f(x-h) = f(x) - hf'(x) + \frac{h^2}{2}f''(x) - \frac{h^3}{6}f'''(c_1), \tag{24}$$

and

$$f(x-2h) = f(x) - 2hf'(x) + \frac{4h^2}{2}f''(x) - \frac{8h^3}{6}f'''(c_2), \tag{25}$$

where $x - h < c_1 < x$ and $x - 2h < c_2 < x$. Substituting the above two equation into $p_2'(x)$, we have

$$\begin{aligned} p_2'(x) &= \frac{f(x-2h) - 4f(x-h) + 3f(x)}{2h} \\ &= \frac{\cancel{f(x)} - 2hf'(x) + \cancel{\frac{4h^2}{2}f''(x)} - \frac{8h^3}{6}f'''(c_2) - 4\left[\cancel{f(x)} - hf'(x) + \cancel{\frac{h^2}{2}f''(x)} - \frac{h^3}{6}f'''(c_1)\right] + \cancel{3f(x)}}{2h} \\ &= f'(x) + h^2\left(-\frac{2f'''(c_2)}{3} + \frac{f'''(c_1)}{3}\right) \end{aligned}$$

Thus, in this case,

$$E = f'(x) - p_2'(x) = h^2\left(\frac{2f'''(c_2)}{3} + \frac{f'''(c_1)}{3}\right) = O(h^2). \tag{26}$$

and the difference formula is second order.

# 4 The method of undetermined coefficients

The method of undetermined coefficients is a procedure used in deriving formulas for numerical differentiation and numerical integration. We describe this method by solving Example 2.

Suppose

$$f'(x) \approx D_h f(x) \equiv Af(x-2h) + Bf(x-h) + Cf(x), \tag{27}$$

where $A, B, C$ are constants to be determined so that $D_h f(x)$ is as accurate as an approximation as possible. Now, from Taylor series,

$$f(x-2h) = f(x) - 2hf'(x) + \frac{4h^2}{2}f''(x) - \frac{8h^3}{6}f'''(c_2) \tag{28}$$

$$f(x-h) = f(x) - hf'(x) + \frac{h^2}{2}f''(x) - \frac{h^3}{6}f'''(c_1) \tag{29}$$

$$f(x) = f(x) \tag{30}$$

If we multiply (28), (29), (30) by $A, B, C$, respectively, and add, then, from (27), we have

$$D_h f(x) = (A+B+C)f(x) + (-2hA - hB)f'(x) + \left(\frac{4h^2}{2}A + \frac{h^2}{2}B\right)f''(x) \tag{31}$$

$$- A\frac{8h^3}{6}f'''(c_2) - B\frac{h^3}{6}f'''(c_1) \tag{32}$$

If

$$D_h f(x) \approx f'(x) \tag{33}$$

for an arbitrary function $f(x)$, we require

$$\begin{aligned} A + B + C &= 0, \\ -2hA - hB &= 1, \\ \left(\frac{4h^2}{2}A + \frac{h^2}{2}B\right) &= 0. \end{aligned} \tag{34}$$

On solving this system, we obtain

$$A = \frac{1}{2h},\ B = -\frac{4}{2h},\ C = \frac{3}{2h}.$$

Then

$$D_h f(x) \equiv \frac{f(x-2h) - 4f(x-h) + 3f(x)}{2h}. \tag{35}$$

## 5 An Application of Numerical Differentiation Formulas

Consider the second order linear two-point boundary value problem,

$$\begin{cases} -u''(x) + \kappa u(x) = f(x), \quad x \in I, \\ u(0) = g_0, \quad u(1) = g_1, \end{cases} \tag{36}$$